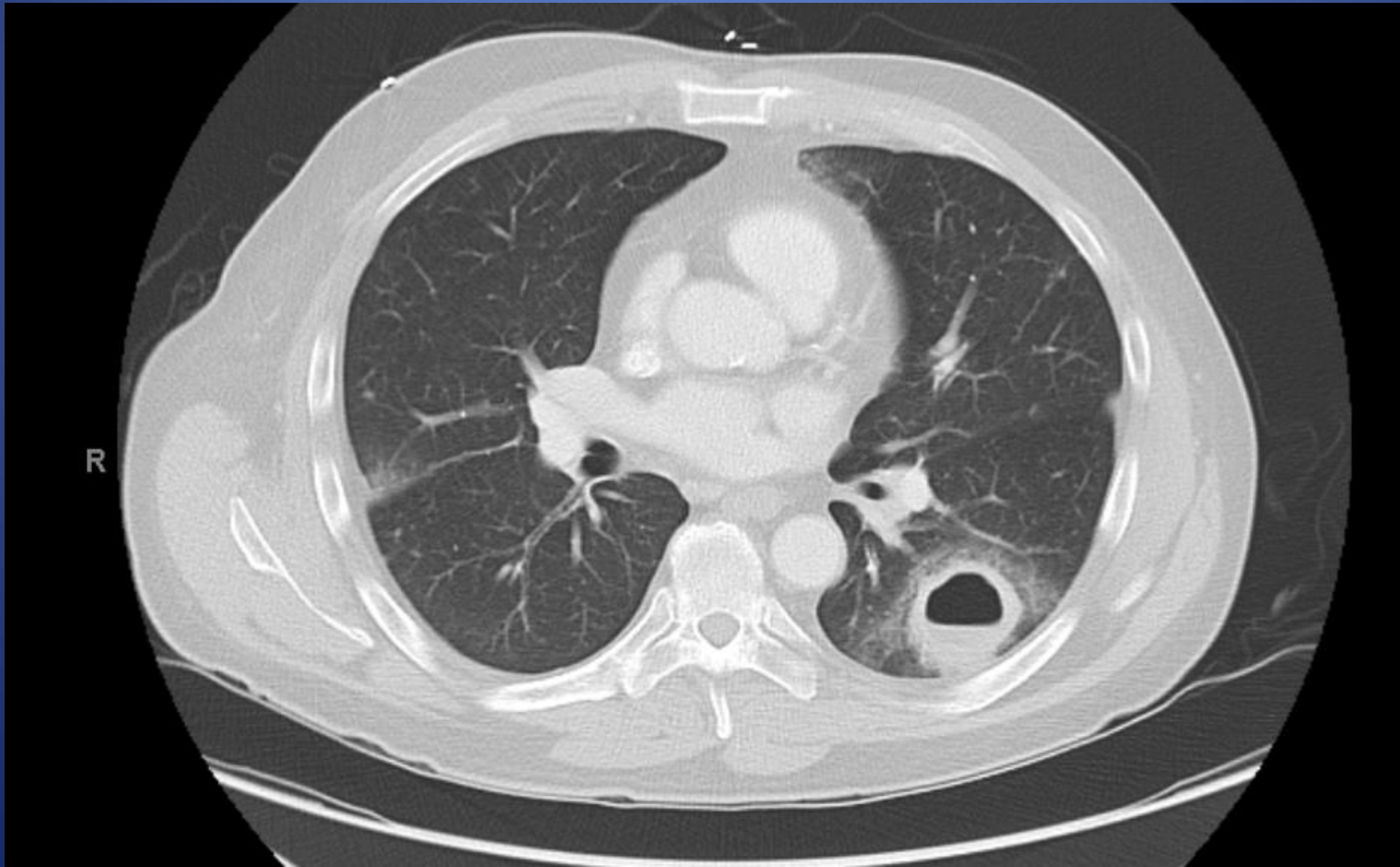


Case 1

- 51 yo man admitted for 2 week history of fevers, night sweats, cough with hemoptysis, malaise. Moderate swelling and stiffness in hands and knees. Some eye redness and photophobia. He works in agriculture in the central valley. No recent travel or ill contacts

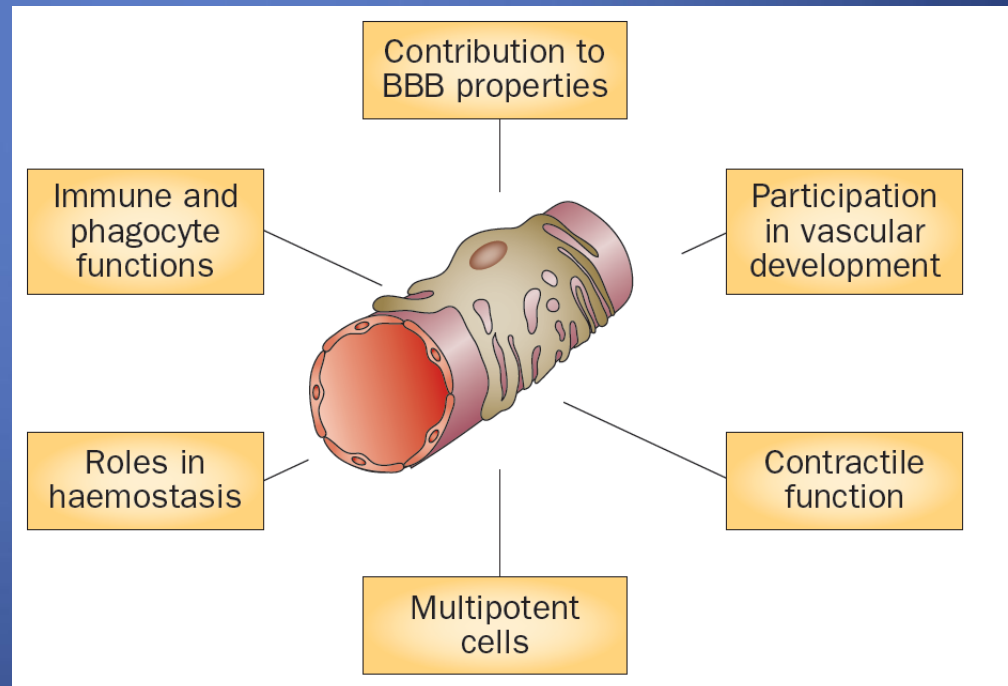


VASCULITIS: OUTLINE

- *Definition*
- *Epidemiology*
- *Classification*
- *Clinical Features*
- *Specific Examples*

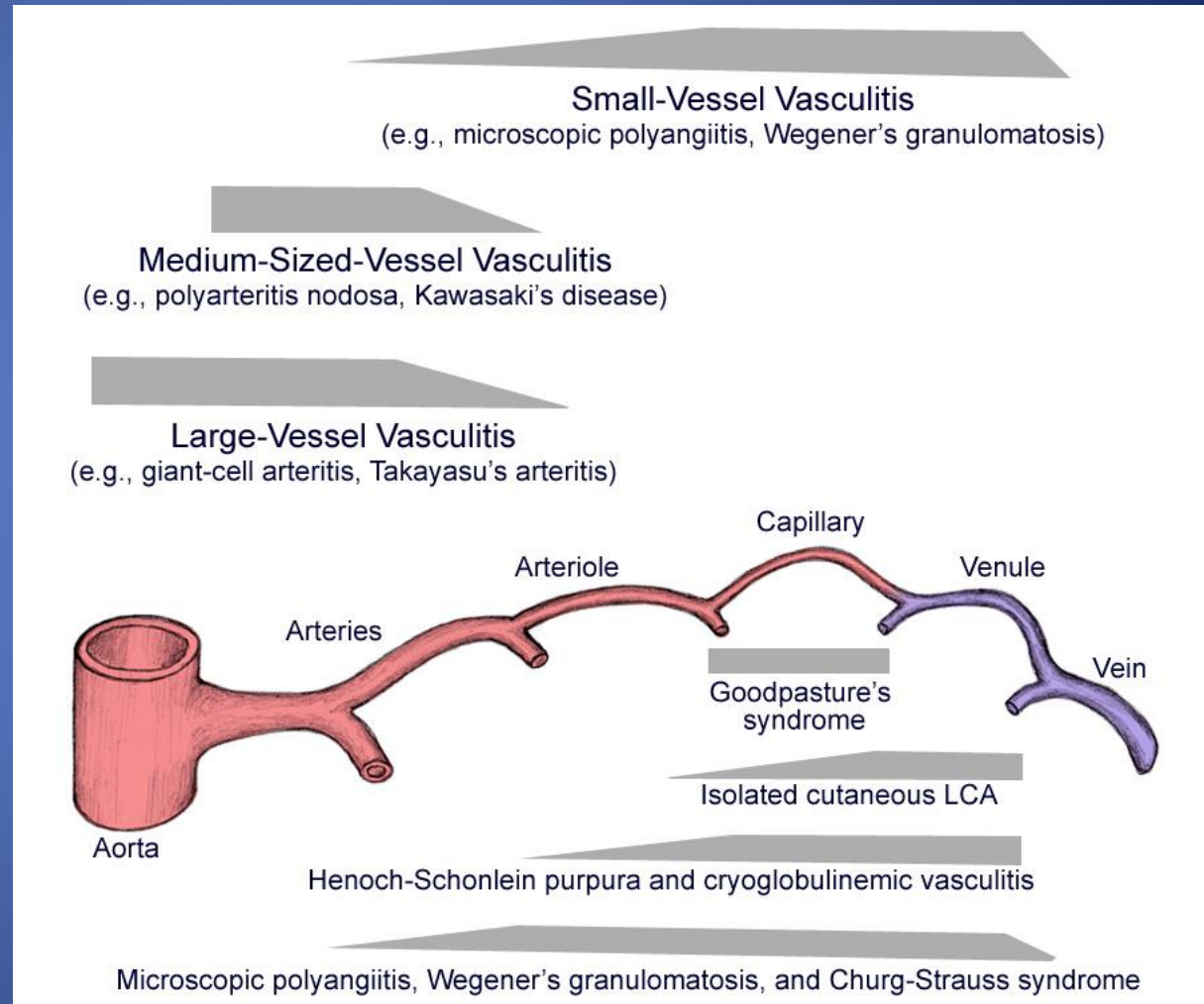
Definition/Pathogenesis

- Heterogeneous set of diseases characterized by inflammatory destruction of blood vessels
 - Immune complex deposition
 - Intimal proliferation
 - Thrombosis
 - Granuloma
 - Tissue necrosis



Approaches to classification of vasculitis

- Primary vs Secondary
- ***Vessel size***
- Laboratory markers



- * Lots of overlap and exceptions come into play

Vasculitis: Classification by vessel size

- Small vessels

- Drug-induced, hypersensitivity
- Henoch-Schönlein purpura
- Cryoglobulinemia
- Antineutrophil cytoplasmic antibody (ANCA) associated
 - Wegener's granulomatosis (GPA)
 - Churg-Strauss syndrome (EGPA)
 - Microscopic polyangiitis (MPA)
- Vasculitis associated with malignancy/infections
- Vasculitis associated with other rheumatic diseases
- Behcet's

- Small, medium arteries

- Polyarteritis nodosa (PAN)
- Rheumatoid vasculitis
- SLE vasculitis
- Buerger's (thromboangiitis obliterans)

- Large arteries

- Giant cell or temporal arteritis (GCA)
- Takayasu arteritis

When to suspect vasculitis: clinical features

- Multisystem disease (MSK, nervous, renal, pulmonary, GI)
- Unexplained constitutional signs and symptoms
 - Weight loss, FUO, anemia
- Skin lesions (palpable purpura)
- Ischemic vascular changes (gangrene, claudication, “Raynaud’s”, livedo)
- Mononeuritis multiplex
- Pain (abdominal - intestinal angina; testicular, arthritis)

Case 2

- 18 yo Caucasian woman presents with 2 wks of abdominal pain, hematochezia, joint swelling – ankles and knees with worsening rash.



Question

A skin biopsy is most likely to show:

- Interface dermatitis
 - Mucin deposition
 - IgA deposition
 - Keratoderma blenorrhagicum
 - Pyoderma gangrenosum
-
- Henoch schonlein purpura (HSP)/IgA vasculitis



Case 3

- 47 yo man with recently diagnosed pancreatic ca presents with dusky purplish tone to tips of fingers and toes in past month – mod aching and numbness.
- SH:
 - ½ ppd x yrs
 - Mod IVDA
- Your exam:.....



- What is the best explanation?
 - acrocyanosis
 - Buerger's (thromboangiitis obliterans)
 - Raynauds
 - cryoglobulinemia
 - chilblains

Cryoglobulinemic vasculitis

- Immunoglobulins undergo reversible precipitation in the cold
 - autoimmune, infectious (HCV), malignancy
- Clinical features
 - Ulceration and necrosis
 - “Raynaud’s” phenomenon
- Clinical features related to immune complex deposition
 - Palpable purpura
 - Arthralgia
 - Nephritis
 - Neuropathy

Cryoglobulinemic Vasculitis: Treatment

Mild disease → Treat underlying cause

Severe disease:

- corticosteroids
- cyclophosphamide
- mycophenolate
- plasmapheresis
- rituximab?



ACR 1990 criteria for classification of polyarteritis nodosa

- Must have at least 3 criteria present.
 - Weight loss > 4 kg
 - Livedo reticularis
 - Testicular pain or tenderness
 - Myalgias, weakness, or leg tenderness
 - Mononeuropathy or polyneuropathy
 - Elevated BUN/creatinine
 - Hepatitis B virus
 - Biopsy of small or medium artery containing PMN

THINK: “MULTIPLE ORGAN SYSTEMS” (GI, RENAL, SKIN, NERVOUS, HEME)

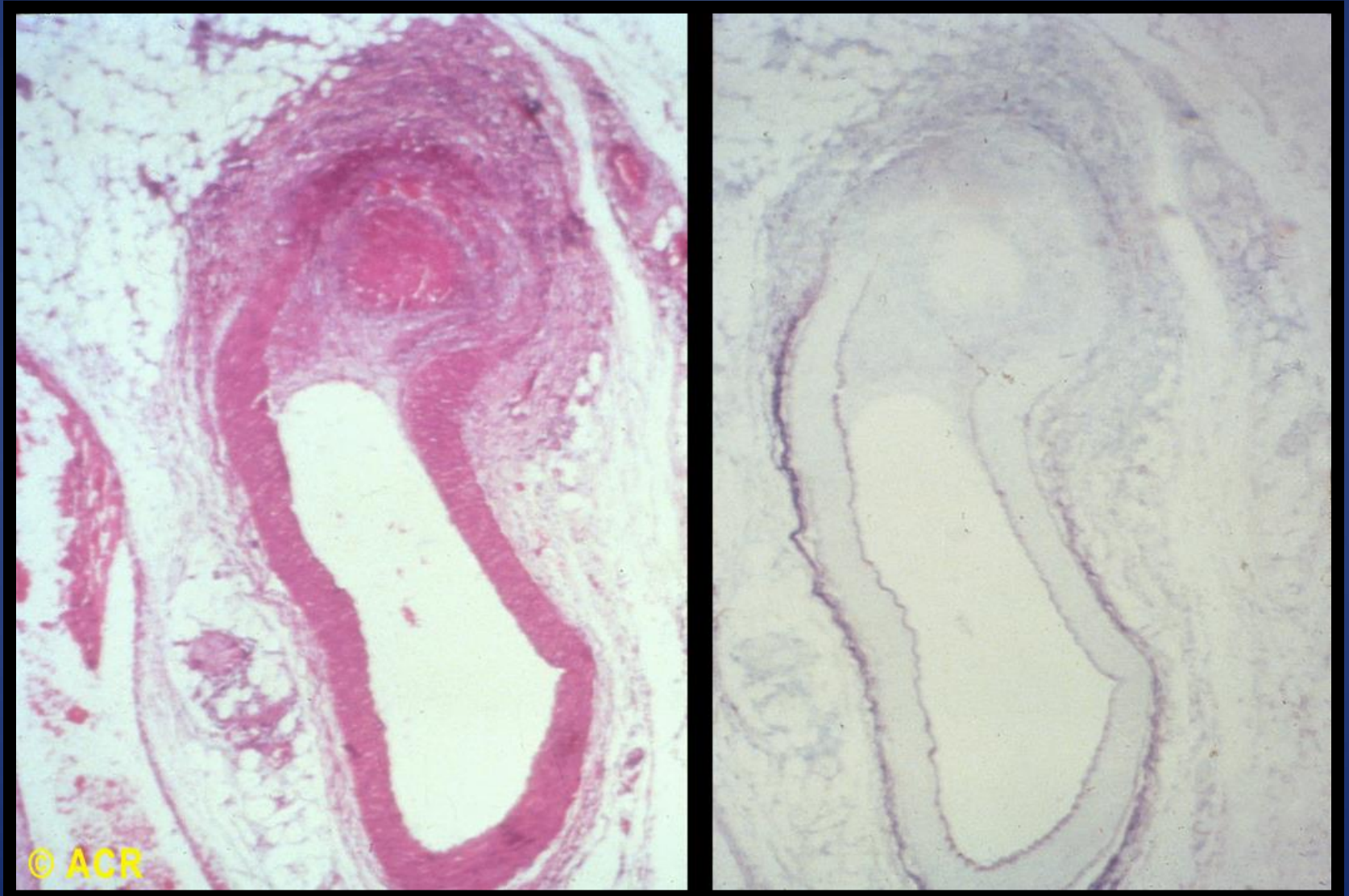
Livedo reticularis



© ACR



Polyarteritis nodosa: necrotizing arteritis (photomicrographs)



Polyarteritis nodosa (mesenteric angiogram)



PAN: Treatment

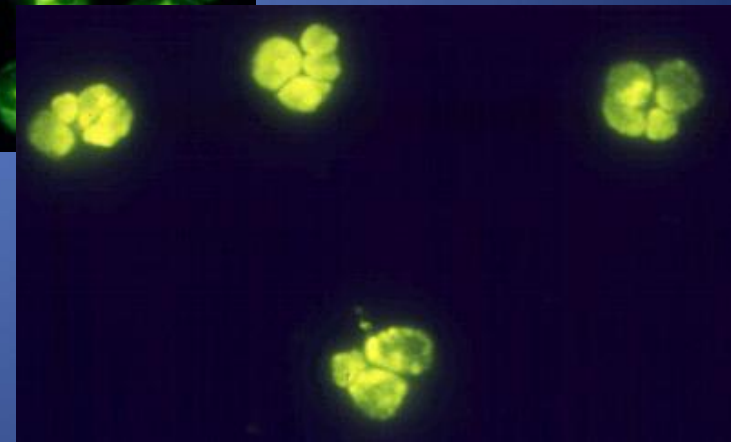
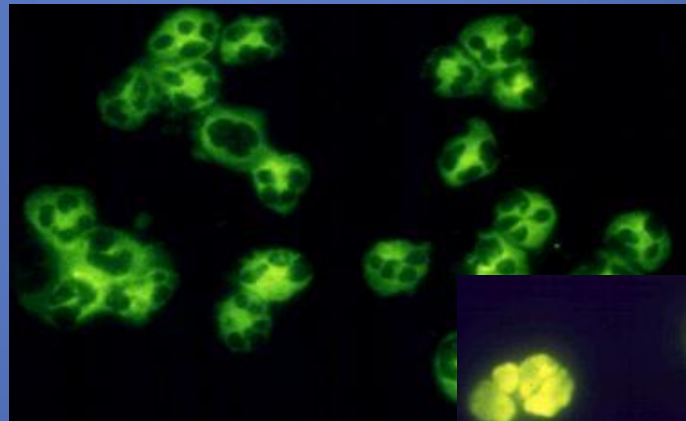
- Idiopathic:
 - Prednisone 1 mg/kg/day
 - Pulse steroids, if severe
 - Cyclophosphamide if steroid refractory
- HBV related:
 - Steroids
 - Plasma exchange
 - Anti-viral therapy

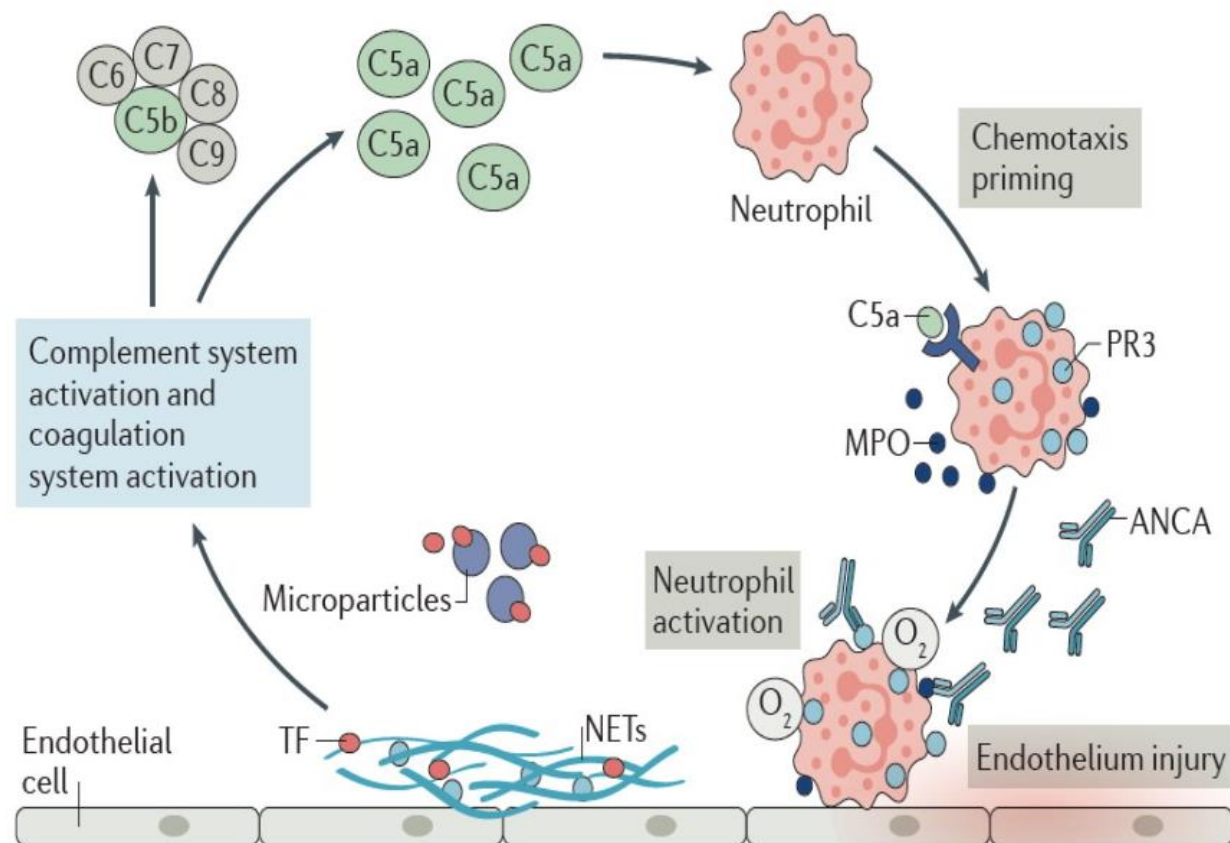
ACR classification criteria: Wegener's Granulomatosis/ Granulomatosis with Polyangiitis (as of Jan 2011)

- Must have at least 2 of the 4 criteria present:
 - Nasal/oral inflammation (oral ulcers or bloody nasal drainage)
 - Abnormal chest radiograph (nodules, fixed infiltrates, cavities)
 - Urinary sediment (>5 RBC/hpf or RBC casts) with glomerulonephritis
 - Granulomatous inflammation and Giant Cells on biopsy (in wall of artery or arteriole, perivascular, or extravascular)

WG/GPA

- Clinical presentation
 - Initially upper airway, sinus/ear manifestations
 - Refractory symptoms, pulmonary and renal usually follow
- Lab studies
 - C-ANCA, anti-proteinase-3 (microscopic polyangiitis (MPA) is P-ANCA, anti-myeloperoxidase)
 - ESR/CRP
 - Anemia
 - renal
- Treatment
 - Corticosteroids
 - Cyclophosphamide (PO or IV)
 - Azathioprine, mycophenolate
 - Rituximab (April 11, 2011)
 - Avacopan (2021)





Chen M et al. *Nat Rev Nephrol* 2017

ESTABLISHED IN 1812

FEBRUARY 18, 2021

VOL. 384 NO. 7

Avacopan for the Treatment of ANCA-Associated Vasculitis

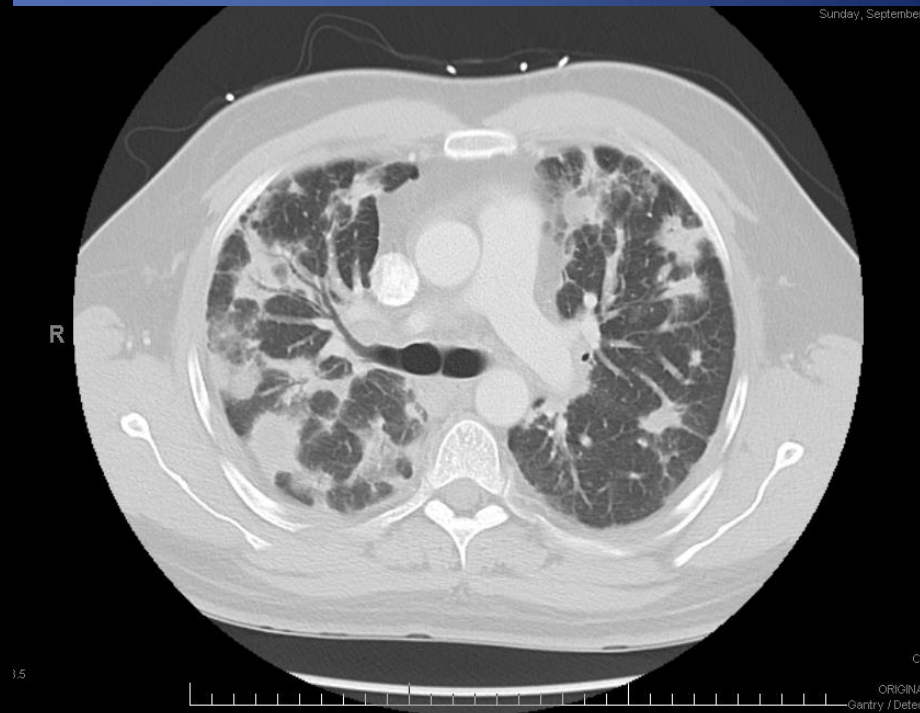
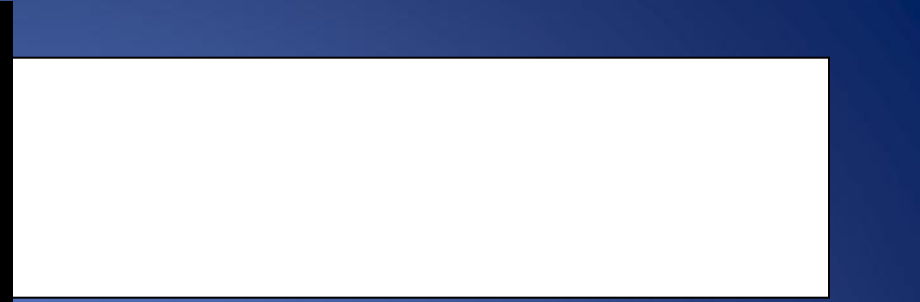
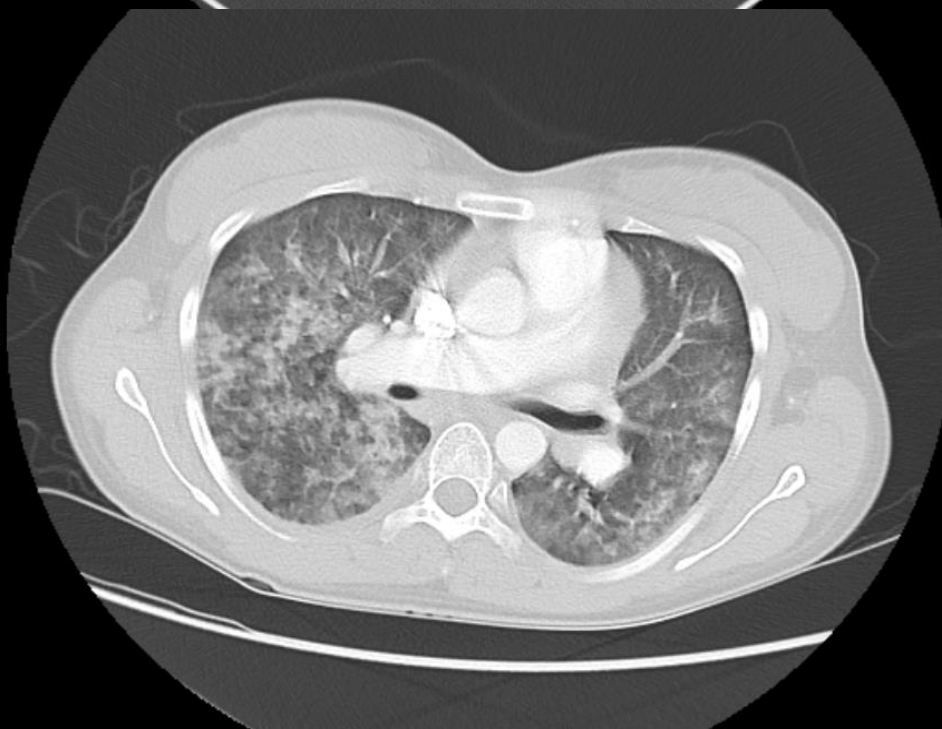
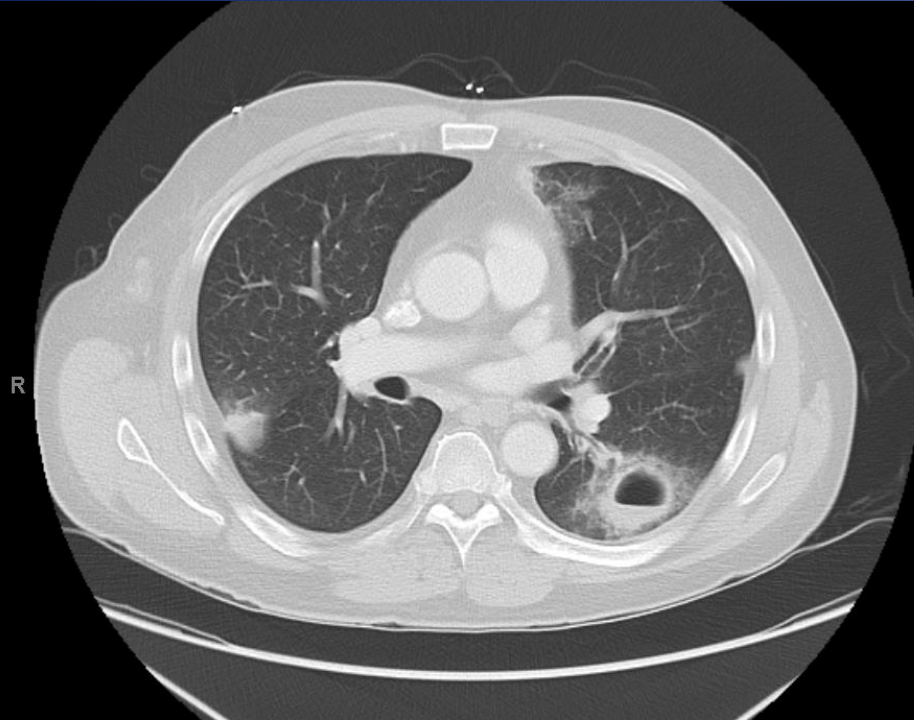
David R.W. Jayne, M.D., Peter A. Merkel, M.D., M.P.H., Thomas J. Schall, Ph.D., and Pirow Bekker, M.D, Ph.D.,
for the ADVOCATE Study Group*

WG/GPA: saddle-nose deformity



Wegener's granulomatosis: lungs (radiograph)







Case 4

- 59 yo woman with asthma x 5 yrs admitted with SOB and nonproductive cough and rapid weight loss – 20 lbs in past 2 wks with anorexia, mod arthralgias in ankles and knees.
- SH: no tob/etoh/drugs, no recent travel
- Married accountant
- CXR: diffuse patchy infiltrates cw multilobar pneumonia
- What lab is most likely to yield diagnostic criteria:
 - ESR/CRP
 - CBC with diff
 - ANCA
 - Serum IGE

Component	Latest Ref Rng	5/17/2010 	5/18/2010	5/19/2010	8/26/2010
WBC'S AUTO	4.0 - 11.0 thou/mcl	20.2 (H)	20.2 (H)	17.6 (H)	
RBC AUTO	4.2 - 5.4 mil/mcl	3.65 (L)	3.86 (L)	3.70 (L)	
HGB	12.0 - 16.0 g/dl	11.2 (L)	12.2	11.6 (L)	
HCT AUTO	37 - 47 %	32.7 (L)	34.9 (L)	32.7 (L)	
MCV	81 - 99 fl	89.6	90.4	88.3	
MCH	27 - 35 pg/cell	30.6	31.7	31.3	
MCHC	32 - 37 g/dl	34.1	35.0	35.4	
RDW, BLOOD	11.5 - 14.5 %	14.1	14.3	14.3	
PLT'S AUTO	130 - 400 thou/mcl	289	317	329	
MPV	7.4 - 10.4 fl	7.9	7.9	7.3 (L)	
LYMPHS % AUTO	20 - 51 %	10.5 (L)	10.2 (L)	9.5 (L)	
MONOS %	1 - 12 %	4.4	3.4	6.0	
NEUTROPHILS % AUTO	42 - 75 %	44.2	44.4	79.5 (H)	
EOS % AUTO	0 - 10 %	40.8 (H)	41.9 (H)	4.9	
BASO'S % AUTO	0 - 1 %	0.1	0.1	0.1	
IGE, QL	<114 KU/L	637.00 (H)			59.70

1. VASCULITIS, MULTIFOCAL (ADIPOSE AND SKELETAL MUSCLE) WITH: INVOLVEMENT OF SMALL AND MEDIUM CALIBER VESSELS, VENULES AND CAPILLARIES BY INFLAMMATION, PREDOMINANTLY EOSINOPHILIC

2. INTRAVASCULAR EOSINOPHILS SUGGESTING PERIPHERAL HYPEREOSINOPHILIA

COMMENT: The clinical history and changes seen in this biopsy are consistent with the clinical impression of Churg-Strauss syndrome though a definitive diagnosis requires clinical and serologic correlation. While inflammation predominantly involves blood vessels including endomysial capillaries, a minor component of myositis is may be present.

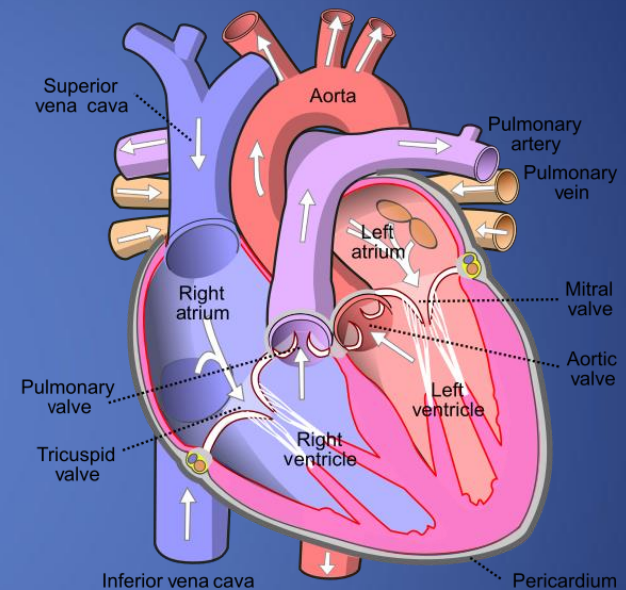
Component	Latest Ref Rng	5/17/2010 	5/18/2010	5/19/2010	8/26/2010
WBC'S AUTO	4.0 - 11.0 thou/mcl	20.2 (H)	20.2 (H)	17.6 (H)	
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PLT'S AUTO	130 - 400 thou/mcl	289	317	329	
MPV	7.4 - 10.4 fl	7.9	7.9	7.3 (L)	
LYMPHS % AUTO	20 - 51 %	10.5 (L)	10.2 (L)	9.5 (L)	
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NEUTROPHILS % AUTO	42 - 75 %	44.2	44.4	79.5 (H)	
EOS % AUTO	0 - 10 %	40.8 (H)	41.9 (H)	4.9	
BASO'S % AUTO	0 - 1 %	0.1	0.1	0.1	
IGE, QL	<114 KU/L	637.00 (H)			59.70

1. VASCULITIS, MULTIFOCAL (ADIPOSE AND SKELETAL MUSCLE) WITH: INVOLVEMENT OF SMALL AND MEDIUM CALIBER VESSELS, VENULES AND CAPILLARIES BY INFLAMMATION, PREDOMINANTLY **EOSINOPHILIC**
2. INTRAVASCULAR EOSINOPHILS SUGGESTING PERIPHERAL HYPEREOSINOPHILIA

COMMENT: The clinical history and changes seen in this biopsy are consistent with the clinical impression of Churg-Strauss syndrome though a definitive diagnosis requires clinical and serologic correlation. While inflammation predominantly involves blood vessels including endomysial capillaries, a minor component of myositis is may be present.

Eosinophilic granulomatosis with polyangiitis (EGPA)(Churg Strauss)

- 4 of the 6 criteria:
 - Asthma
 - Peripheral Eosinophilia > 10%
 - Neuropathy, mono or poly
 - Pulmonary infiltrates, non-fixed
 - Paranasal sinus abnormality
 - Extravascular eosinophils



“Wegener’s with asthma, eosinophilia”

Treatments are similar except IL-5 targets (benralizumab and mepolizumab)

ORIGINAL ARTICLE

Mepolizumab or Placebo for Eosinophilic Granulomatosis with Polyangiitis

Michael E. Wechsler, M.D., Praveen Akuthota, M.D., David Jayne, F.Med.Sci., Paneez Khoury, M.D., M.H.S., Amy Klion, M.D., Carol A. Langford, M.D., M.H.S., Peter A. Merkel, M.D., M.P.H., Frank Moosig, M.D., Ulrich Specks, M.D., Maria C. Cid, M.D., Raashid Luqmani, D.M., Judith Brown, Ph.D., et al., for the EGPA Mepolizumab Study Team*

[Article](#) [Figures/Media](#)

[Metrics](#)

May 18, 2017

N Engl J Med 2017; 376:1921-1932

Polymyalgia rheumatica (PMR): clinical and laboratory features

- Pelvic and shoulder girdle aching
- Morning stiffness
- Anemia
- Elevated ESR and C-reactive protein
- Rapid response to low doses of corticosteroids

PMR: epidemiology and treatment

- Onset after age 50 and usually after 60
- Highest incidence in individuals of N. Europe
- Female:male ratio 2:1
- Incidence approaches 1% in older populations
- Initial **10-20 mg prednisone** daily x 1 month → immediate results
 - Gradually taper – 1 mg less per month
 - Methotrexate and interleukin 6....

February 28, 2023 at 8:00 PM EST

« Back

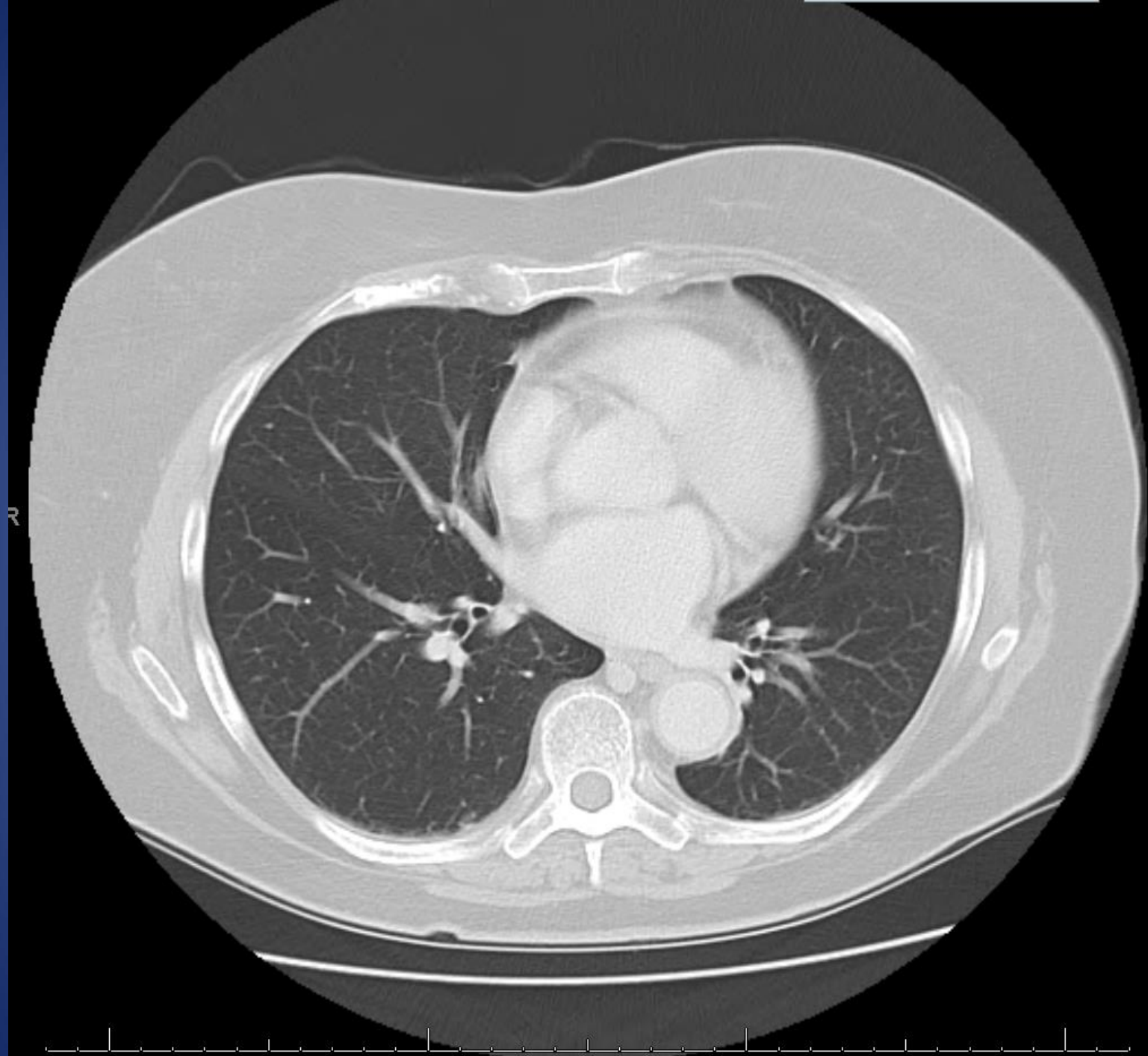


KEVZARA® (SARILUMAB) APPROVED BY FDA AS FIRST AND ONLY BIOLOGIC INDICATED FOR PATIENTS WITH POLYMYALGIA RHEUMATICA

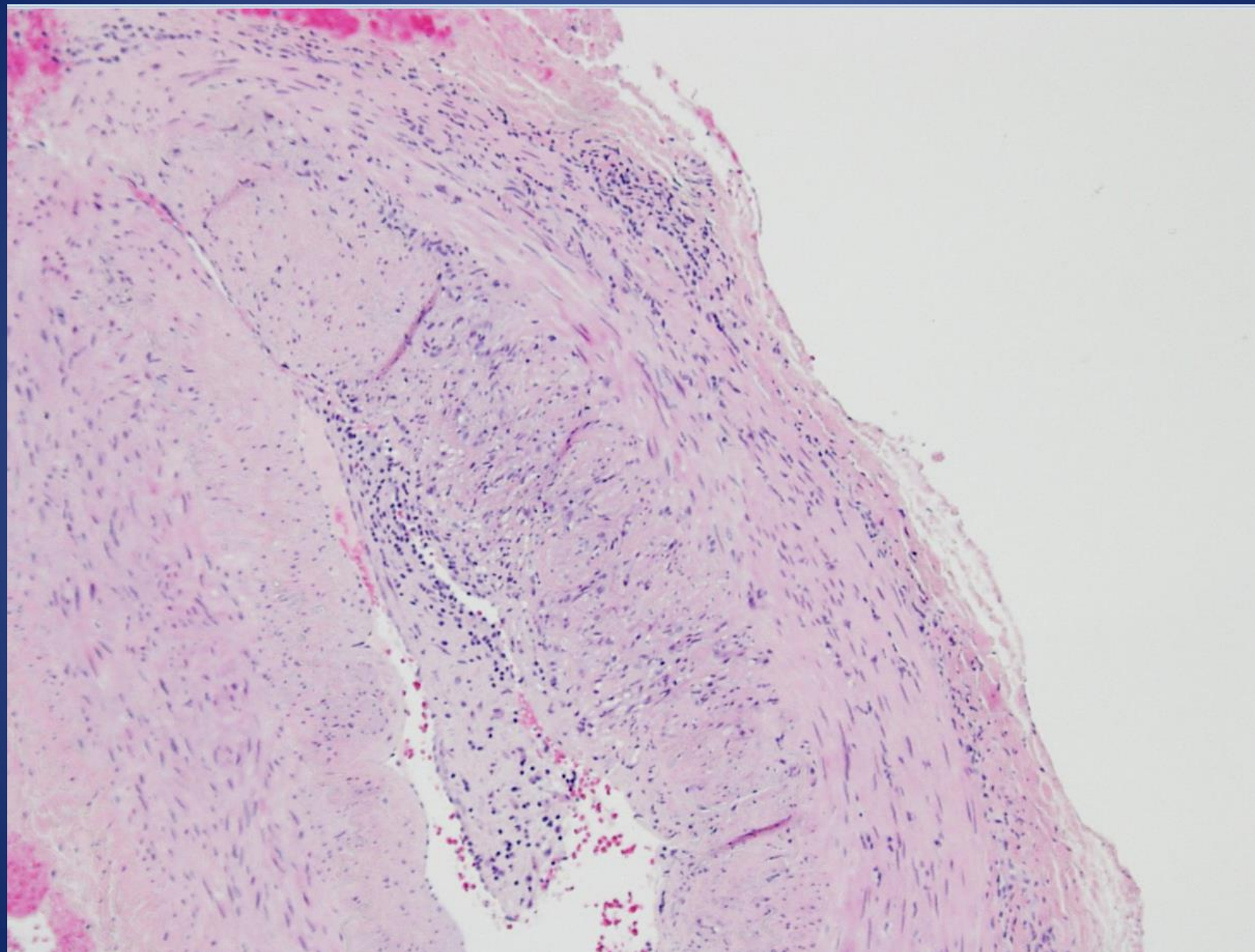
ACR 1990 criteria for the classification of giant cell arteritis (GCA) or temporal arteritis

Must have at least 3 of the 5 criteria present:

- Age > 50 years at disease onset
- New headache
- Temporal artery abnormality (tender or decreased pulse)
- Elevated Westergren ESR > 50 mm/hr
- Abnormal artery biopsy: mononuclear cell infiltrate, granulomatous inflammation, usually multinucleated giant cells







GCA-other clinical features

- Clinical features
 - Jaw claudication
 - Visual loss; diplopia
 - PMR symptoms
 - Weight loss, fever
- Laboratory features
 - Elevated ESR/CRP
 - Anemia

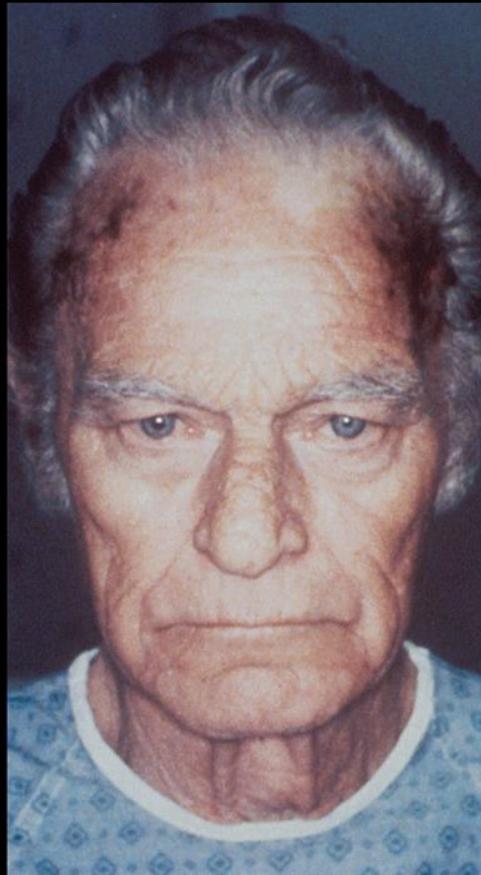
Relationship of polymyalgia (PMR) to giant cell arteritis (GCA)

- 40-60% of patients with GCA have PMR symptoms
- 10-15% of patients with PMR can develop GCA
- PMR symptoms can occur before, with, or after GCA symptoms in patients with GCA
- Treatment of GCA requires **larger doses** of corticosteroids than does treatment of PMR
 - Role of interleukin 6

GCA: temporal artery



Giant cell arteritis: scalp necrosis



GCA: retinal ischemia



GCA - treatment

- Prednisone 1 mg/kg/day (60 mg) x 30 days, then taper
- IV methylprednisolone if severe (vision loss)
- Glucocorticoid induced osteoporosis (GIO)
 - Calcium with vit D
 - Bisphosphonates
 - Teriparatide
- Targeted therapies
 - Tocilizumab (anti-interleukin 6)
 - Upadacitinib (janus kinase inhibitor)

Component	ESR	CRP
Latest Ref Rng & Units	0 - 30 mm/Hr	<=7.4 mg/L
10/31/2016  	100 (H)	55.8 (H) 
11/18/2016  	71 (H)	18.5 (H) 
1/13/2017  	52 (H)	30.5 (H) 
2/21/2017  	51 (H)	28.2 (H) 
3/10/2017  	51 (H)	24.6 (H) 
3/22/2017  	68 (H)	59.3 (H) 
5/4/2017  	55 (H)	28.0 (H) 
6/6/2017  	43 (H)	24.9 (H) 

Component	ESR	CRP
Latest Ref Rng & Units	0 - 30 mm/Hr	<=7.4 mg/L
10/31/2016 	100 (H)	55.8 (H)
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3/10/2017 	51 (H)	24.6 (H)
3/22/2017 	68 (H)	59.3 (H)
5/4/2017 	55 (H)	28.0 (H)
6/6/2017 	43 (H)	24.9 (H)
6/29/2017 	9	<1.0
9/5/2017 	5	<1.0
10/23/2017 	4	<1.0
12/14/2017 	5	<1.0
1/23/2018 	4	<1.0

FDA approves first drug to specifically treat giant cell arteritis

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JULY 27, 2017

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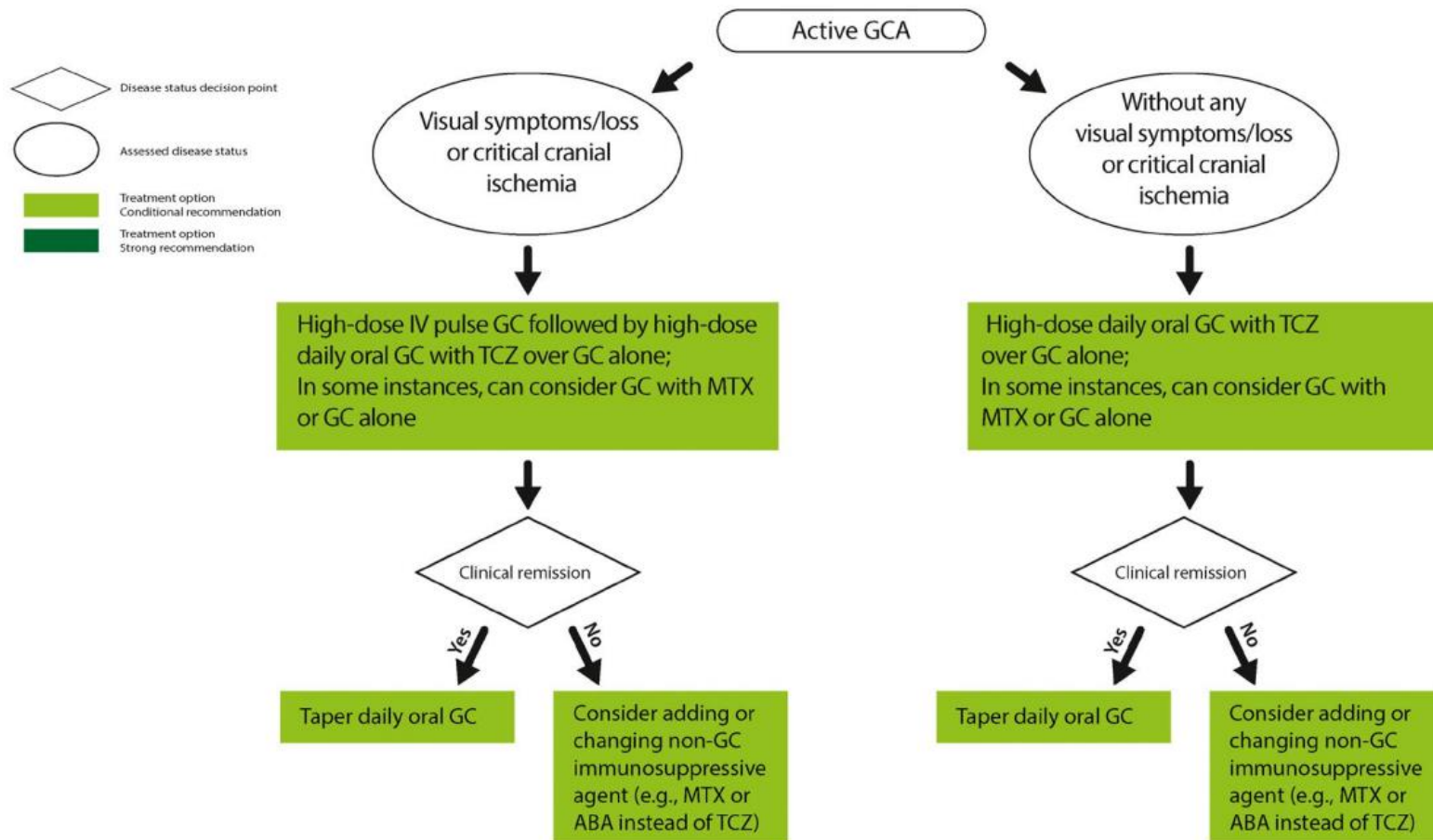
For Immediate
Release

Trial of Tocilizumab in Giant-Cell Arteritis

J.H. Stone, K. Tuckwell, S. Dimonaco, M. Klearman, M. Aringer, D. Blockmans, E. Brouwer, M.C. Cid, B. Dasgupta, J. Rech, C. Salvarani, G. Schett, H. Schulze-Koops, R. Spiera, S.H. Unizony, and N. Collinson

2021 American College of Rheumatology/Vasculitis Foundation Guideline for the Management of Giant Cell Arteritis and Takayasu Arteritis

Overview of treatment of giant cell arteritis (GCA)



ABA = abatacept, AZA = azathioprine, GC = glucocorticoids, IV = intravenous, MTX = methotrexate, TCZ = tocilizumab

Figure 1. Overview of treatment of giant cell arteritis.

GCA 2025

The NEW ENGLAND JOURNAL of MEDICINE

A Phase 3 Trial of Upadacitinib for Giant-Cell Arteritis

A Research Summary based on Blockmans D et al. | 10.1056/NEJMoa2413449 | Published on April 2, 2025

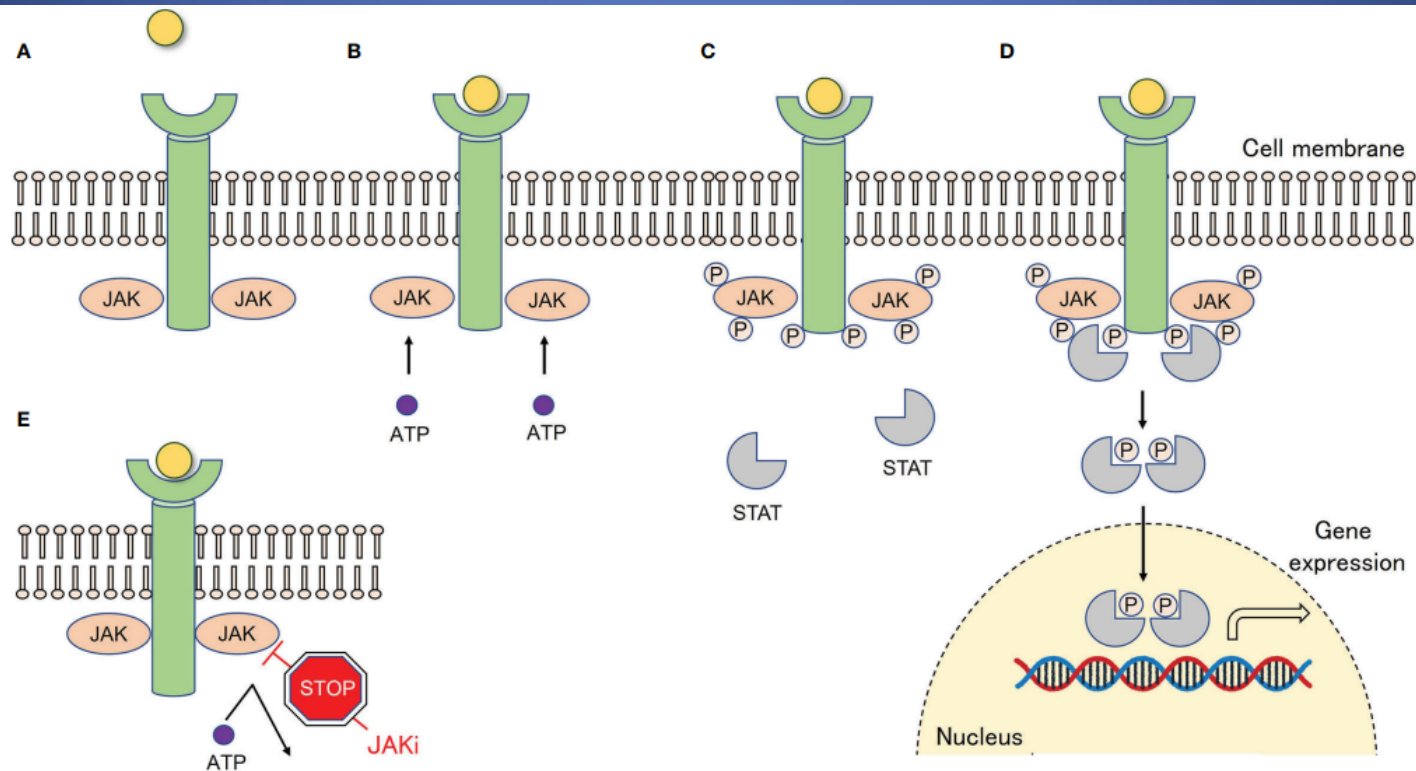


FIGURE 1 | The JAK-STAT pathway and mechanism of action of JAK inhibitors. **(A)** Type I and type II cytokines utilize the Janus kinase (JAK)-signal transducer and activation of transcription (STAT) pathway. **(B)** When type I and II cytokines bind to their receptors on the cell surface, JAKs bound to the intracellular domains are phosphorylated by adenosine triphosphate binding. **(C)** Phosphorylated JAKs, in turn, phosphorylate the receptor end. **(D)** The transcription factor STAT binds to the receptor end, and phosphorylated STATs form a dimer, which is then transferred to the nucleus to regulate gene expression. **(E)** JAK inhibitors competitively bind to the binding site of ATP, inhibiting phosphorylation of JAK and exerting their effects. ATP, Adenosine triphosphate; JAK, Janus kinase; JAKi, Janus kinase inhibitor; P, Phosphate; STAT, Signal transducer and activator of transcription.

Final Case

21 year old Japanese woman complains of 2 months of worsening arm fatigue with use (L>R) and dizziness and occasional near syncope; 10 lbs wt loss unintentionally.

ROS: no vision changes, mild intermittent fevers

PMH: none

PSH: none

SH: no tob or etoh or drugs

FH: noncontributory

EXAM: NAD, alert

clear lungs

soft murmur

no rashes

diminished pulse on L radial

- What is the next most appropriate test:
 - ANA
 - ESR/CRP
 - Temporal artery biopsy
 - Muscle biopsy
 - angiography



5/15/2013			30.1 (H)		57 (H)
7/24/2013			70.8 (H)		74 (H)
11/1/2013			136.2 (H)		118 (H)
2/7/2014			26.4 (H)		53 (H)
6/29/2015			98.3 (H)		>120 (H)
7/28/2015			14.5 (H)		35 (H)
9/8/2015			104.2 (H)		66 (H)



5/15/2013			30.1 (H)		57 (H)
7/24/2013			70.8 (H)		74 (H)
11/1/2013			136.2 (H)		118 (H)
2/7/2014			26.4 (H)		53 (H)
6/29/2015			98.3 (H)		>120 (H)
7/28/2015			14.5 (H)		35 (H)
9/8/2015			104.2 (H)		66 (H)
11/10/2015			1.2		13
1/7/2016			0.4		11
3/7/2016			0.3		12

Takayasu's arteritis

- Rare, large vessel
- Younger, female>male; Asian/Latin
- Claudication, CNS, htn
- Spectrum of GCA?

Treatments:

- steroids
- immunosuppressants
- tnf inhibitors?
- interleukin 6 inhibitor
- vascular surgery



Summary

- *Henoch Schonlein purpura/IgA vasculitis*
- *Cryoglobulinemic vasculitis*
- *Polyarteritis nodosa*
- *Wegener's/GPA, MPA*
- *Churg Strauss/EGPA*
- *PMR/GCA*
- *Takayasu's arteritis*